Lecture 3

# Which momentum? Self-adjointness is a choice

**Objectives.** *The rule $-i\hbar\psi'$ becomes different operators on different domains. On the circle it is self-adjoint. In the box the Dirichlet momentum is only symmetric, and its self-adjoint extensions are the momenta $P_\theta$. We compare their readings and relate finite mean energy to the Dirichlet domain.*

- Use the boundary form to test symmetry.
- Compare the $P_0$ and $P_\pi$ readings of one box state.
- Explain why no self-adjoint extension of $P_D$ squares to the box energy.

## 3.1 From the ring to the circle

Take the ring of *Quantum Mechanics I*, Lecture 8: $N$ sites, the shift $T$ with $Te_x = e_{x+1}$, and the Hamiltonian $c\mathbb{1} - \kappa(T + T^\dagger)$, whose eigenvectors are the Fourier modes $f_k$, with the levels $c - 2\kappa\cos(2\pi k/N)$. The circle's functions $f_k$ below, sampled at the sites, are these modes up to normalisation, so they share the letter. As for the chain of § 1.1, put $c = 2\kappa$, the spacing $a = L/N$ and $\kappa = \hbar^2/2ma^2$. The ring now has the circumference $L$, and its levels are

$$4\kappa \sin^2 \frac{\pi k}{N} = \frac{2\hbar^2 N^2}{mL^2} \sin^2 \frac{\pi k}{N} \longrightarrow \frac{p_k^2}{2m}, \qquad p_k = \frac{2\pi\hbar k}{L},$$

at each fixed integer $k$ as $N \to \infty$, from below. In units of the box's $E_1 = \hbar^2\pi^2/2mL^2$ these are $p_k^2/2m = 4k^2E_1$: the levels 0, then $4E_1$ twice ($k = \pm 1$), $16E_1$ twice, and so on. They are not the box's $E_1, 4E_1, 9E_1, \ldots$. The ring's limit is a particle on a circle.

A particle on a circle of circumference $L$ has a position $x$ in $[0, L)$, where $x = L$ is the point $x = 0$ again. Its wavefunctions form the same space $L^2(0, L)$, with the same inner product, and the functions

$$f_k(x) = \frac{e^{2\pi i k x/L}}{\sqrt{L}} \qquad (k \in \mathbb{Z})$$

form a complete orthonormal basis. This is the Fourier basis of *Linear Algebra*, Ex. 13.25, rescaled, and a standing fact. Write $c_k = \langle f_k, \psi\rangle$.

**Definition 3.1** (The momentum on a circle)**.** The momentum $P$ on the circle has the domain and the action

$$D(P) = \Big\{\psi \in L^2(0, L) : \sum_{k\in\mathbb{Z}} k^2 |c_k|^2 < \infty\Big\}, \qquad P\psi = \sum_{k\in\mathbb{Z}} p_k c_k f_k.$$

**Proposition 3.2** (The momentum on a circle)**.** *The momentum $P$ is densely defined and self-adjoint, and its eigenvalues are exactly the $p_k$, and the eigenvectors for $p_k$ are the nonzero multiples of $f_k$. Every continuously differentiable $\psi$ on $[0, L]$ with $\psi(L) = \psi(0)$ lies in $D(P)$, and $P\psi = -\mathrm{i}\hbar\psi'$.*

*Proof.* The steps of the proof of Theorem 1.8 on density, symmetry, the adjoint and the eigenvalues apply word for word, with $f_k$, $p_k$ and $k \in \mathbb{Z}$ in place of $u_n$, $E_n$ and $n \geq 1$, since $p_k^2$ is a multiple of $k^2$ and the $p_k$ are distinct and real. For the rule, let $\psi$ be continuously differentiable with $\psi(L) = \psi(0)$. Integrating by parts,

$$\langle f_k, -\mathrm{i}\hbar\psi' \rangle = -\mathrm{i}\hbar \left[ \overline{f_k}\psi \right]_0^L + \left\langle -\mathrm{i}\hbar f_k', \psi \right\rangle .$$

The bracket is $-\mathrm{i}\hbar(\psi(L) - \psi(0))/\sqrt{L} = 0$, and $-\mathrm{i}\hbar f_k' = p_k f_k$, so the coefficients of $-\mathrm{i}\hbar\psi'$ are $p_k c_k$. By Parseval's identity, $\sum_k p_k^2 \, |c_k|^2 = \hbar^2 \, \|\psi'\|^2 < \infty$, so $\psi \in D(P)$ and $P\psi = -\mathrm{i}\hbar\psi'$. ∎

We take as the energy of the particle on the circle the diagonal operator with the entries $p_k^2/2m$, which is $P^2/2m$. Its entries are the limits of the ring's levels.

**Self-check 3.1**

1. List the three lowest levels of the circle and of the box, with their multiplicities, in units of $E_1$.

## 3.2 The class $\mathcal{A}$ and the boundary form

In the box, what is the momentum? The rule $-\mathrm{i}\hbar\psi'$ needs a derivative and values at the walls, and an element of $L^2(0, L)$ has neither (§ 1.2). Continuous derivatives are too few. The tent $\tau(x) = \min(x, L - x)$ has a corner at $L/2$, and it lies in the domain of the circle's momentum, since its coefficients $c_k$ vanish for even $k \neq 0$ and are $-L^{3/2}/\pi^2 k^2$ for odd $k$, so that $\sum_k k^2 \, |c_k|^2 < \infty$. What is needed is a derivative that is square-integrable.

**Definition 3.3** (The class $\mathcal{A}$)**.** Let $\mathcal{A}$ be the set of $\psi \in L^2(0, L)$ whose class contains a function of the form

$$\psi(x) = a + \int_0^x g(t)\,\mathrm{d}t \qquad (0 \leq x \leq L),$$

with $a \in \mathbb{C}$ and $g \in L^2(0, L)$. We write $\psi' = g$.

They are the absolutely continuous functions with a square-integrable derivative. The following properties make them usable.

1. *Values.* By Cauchy–Schwarz, $|\psi(y) - \psi(x)| \leq \sqrt{y - x} \, \|g\|$ for $x \leq y$, so $\psi$ is continuous. Two continuous functions that differ at a point differ on an interval, so a class contains at most one continuous function. Hence $\psi(0) = a$ and $\psi(L)$ are well defined.
2. *The derivative.* Suppose that $\psi$ has two such forms, with $g$ and $\tilde{g}$. Then both constants are $\psi(0)$, and $d = g - \tilde{g}$ has $\int_0^x d(t)\,\mathrm{d}t = 0$ for every $x$. So $d$ is orthogonal to the indicator of every interval $[0, x]$, hence to every step function. Step functions approximate continuous functions uniformly, and continuous functions are dense in $L^2(0, L)$, since the smooth functions are (the standing fact S4). So $d = 0$, and the derivative $\psi'$ is determined by $\psi$.

3. *Integration by parts.* For $\varphi, \psi \in \mathcal{A}$,

$$\int_0^L \overline{\varphi}\,\psi'\,\mathrm{d}x = \big[\overline{\varphi}\psi\big]_0^L - \int_0^L \overline{\varphi'}\,\psi\,\mathrm{d}x. \tag{3.1}$$

   For continuous $\varphi'$ and $\psi'$ this is calculus. In general, take continuous $g_j \to \psi'$ and $h_j \to \varphi'$ in $L^2(0,L)$, and their primitives $\psi_j = \psi(0) + \int_0^x g_j$ and $\varphi_j = \varphi(0) + \int_0^x h_j$. By the first fact, $\psi_j \to \psi$ and $\varphi_j \to \varphi$ uniformly, and every term of (3.1) converges. The same holds on every interval $[0,y]$ in place of $[0,L]$.

4. *The product rule.* Let $\psi \in \mathcal{A}$ and let $b$ be continuously differentiable on $[0,L]$. Then $\overline{b} \in \mathcal{A}$, and (3.1) on $[0,y]$, with $\varphi = \overline{b}$, gives $b(y)\psi(y) = b(0)\psi(0) + \int_0^y (b'\psi + b\psi')\,\mathrm{d}t$. So $b\psi \in \mathcal{A}$, with $(b\psi)' = b'\psi + b\psi'$.

For example, the tent lies in $\mathcal{A}$: $\tau(x) = \int_0^x g(t)\,\mathrm{d}t$, with $g = 1$ on $[0, L/2]$ and $g = -1$ on $(L/2, L]$. A function with a jump does not, since by the first fact every element of $\mathcal{A}$ has a continuous function in its class. Nor does $\sqrt{x}$, although it is continuous. On every $[\epsilon, L]$ its derivative is $1/2\sqrt{x}$, and by the second fact, applied on $[\epsilon, L]$, so is any $g$ in a form of $\sqrt{x}$. But $\int_\epsilon^L \mathrm{d}x/4x$ grows without bound as $\epsilon \to 0$, so no such $g$ is square-integrable.

**Proposition 3.4** (The boundary form)**.** *For $\varphi, \psi \in \mathcal{A}$,*

$$\langle \varphi, -\mathrm{i}\hbar\psi' \rangle - \langle -\mathrm{i}\hbar\varphi', \psi \rangle = -\mathrm{i}\hbar\big(\overline{\varphi(L)}\psi(L) - \overline{\varphi(0)}\psi(0)\big). \tag{3.2}$$

*Proof.* Multiply (3.1) by $-\mathrm{i}\hbar$, and use $\overline{-\mathrm{i}\hbar\varphi'} = \mathrm{i}\hbar\overline{\varphi'}$. ∎

The rule on all of $\mathcal{A}$ is not symmetric: for $\psi(x) = x$ the number $\langle \psi, -\mathrm{i}\hbar\psi' \rangle = -\mathrm{i}\hbar L^2/2$ is not real. The walls suggest the boundary condition that removes the right side.

**Definition 3.5** (The momentum with walls)**.** The *Dirichlet momentum* $P_D$ has the domain $D(P_D) = \{\psi \in \mathcal{A} : \ \psi(0) = \psi(L) = 0\}$ and the action $P_D\psi = -\mathrm{i}\hbar\psi'$.

**Proposition 3.6** (The momentum with walls is symmetric)**.** *The Dirichlet momentum $P_D$ is densely defined and symmetric.*

*Proof.* Each $u_n$ lies in $D(P_D)$, so the finite combinations of the $u_n$ do, and they are dense. For $\varphi, \psi \in D(P_D)$, the right side of (3.2) vanishes. ∎

The momentum $P_D$ is what physics texts call the Hermitian momentum of the box.

**Self-check 3.2**

1. Is $\psi(x) = x(L-x)$ in $D(P_D)$? Compute $\langle \psi, P_D\psi \rangle$.
2. Which of these lie in $\mathcal{A}$: the indicator of $[0, L/2]$, the function $x^{3/4}$, the tent $\tau$? Which lie in $D(P_D)$?

## 3.3 The adjoint of the momentum with walls

Recall that $A \subseteq B$ means that $B$ extends $A$ (Definition 0.4), and that adjoints reverse inclusions (Proposition 0.5).

**Proposition 3.7** (A large adjoint)**.** *Every $\varphi \in \mathcal{A}$ lies in $D(P_D^\dagger)$, with $P_D^\dagger\varphi = -\mathrm{i}\hbar\varphi'$. Every complex number $z$ is an eigenvalue of $P_D^\dagger$, with the eigenvector $e^{\mathrm{i}zx/\hbar}$. The operator $P_D$ has no eigenvalue, and it is not self-adjoint.*

*Proof.* Let $\varphi \in \mathcal{A}$ and $\psi \in D(P_D)$. Since $\psi(0) = \psi(L) = 0$, (3.2) gives $\langle \varphi, P_D\psi\rangle = \langle -i\hbar\varphi', \psi\rangle$. By Definition 0.2, with $\chi = -i\hbar\varphi'$, the vector $\varphi$ lies in $D(P_D^\dagger)$. The function $e^{izx/\hbar}$ lies in $\mathcal{A}$, and $-i\hbar(e^{izx/\hbar})' = ze^{izx/\hbar}$. If $-i\hbar\psi' = z\psi$ for some $\psi \in \mathcal{A}$, then by the product rule $(e^{-izx/\hbar}\psi)' = e^{-izx/\hbar}(\psi' - iz\psi/\hbar) = 0$, so $e^{-izx/\hbar}\psi$ is the constant $\psi(0)$, and $\psi = \psi(0)\,e^{izx/\hbar}$. In $D(P_D)$, $\psi(0) = 0$ and $\psi = 0$. Finally, the constant 1 lies in $D(P_D^\dagger)$ and not in $D(P_D)$. ∎

The converse is a cited theorem.

**Theorem 3.8** (The adjoint · *cited*)**.** *$D(P_D^\dagger) = \mathcal{A}$. So $P_D^\dagger$ is the rule $-i\hbar\varphi'$ with no boundary condition.*

Teschl, (2.32), proves it for $-i\,d/dx$ on the continuously differentiable functions on $[0, 2\pi]$ that vanish at both ends; a change of scale carries it to $[0, L]$ and $\hbar$. Call that operator $A_0$. Since $A_0 \subseteq P_D$, adjoints reverse the inclusion: $P_D^\dagger \subseteq A_0^\dagger$, and $A_0^\dagger$ is the rule on $\mathcal{A}$. So $P_D$ is too small, and its adjoint is too large: $P_D$ has no eigenvalue, and $P_D^\dagger$ has every complex number as one. The symmetric operator $P_D$ is not an observable.

**Self-check 3.3**

1. Give an eigenvector of $P_D^\dagger$ with the eigenvalue $i\hbar/L$. Why is it not an eigenvector of $P_D$?

## 3.4 The momenta $P_\theta$

A self-adjoint momentum $P$ in the box with $P_D \subseteq P$ satisfies $P \subseteq P_D^\dagger$, since adjoints reverse inclusions and $P = P^\dagger$. So it is the rule $-i\hbar\psi'$ on a domain between $D(P_D)$ and $\mathcal{A}$. Two functions of $\mathcal{A}$ with the same values at the walls differ by an element of $D(P_D)$, so the domain is fixed by a condition on the boundary values. By (3.2) with $\varphi = \psi$, the mean $\langle \psi, -i\hbar\psi'\rangle$ is real only if $|\psi(L)| = |\psi(0)|$.

**Definition 3.9** (The momenta $P_\theta$)**.** For $\theta \in [0, 2\pi)$, the momentum $P_\theta$ has the domain

$$D(P_\theta) = \left\{\psi \in \mathcal{A} :\ \psi(L) = e^{i\theta}\psi(0)\right\}$$

and the action $P_\theta\psi = -i\hbar\psi'$.

**Theorem 3.10** (Self-adjoint momenta in a box · *sketched*)**.** *For each $\theta \in [0, 2\pi)$, the momentum $P_\theta$ is self-adjoint, and $P_D \subseteq P_\theta \subseteq P_D^\dagger$. The functions*

$$e_{\theta,k}(x) = \frac{e^{i(\theta+2\pi k)x/L}}{\sqrt{L}} \qquad (k \in \mathbb{Z})$$

*form a complete orthonormal basis of eigenvectors of $P_\theta$, with the eigenvalues $\hbar(\theta + 2\pi k)/L$. The momentum $P_0$ is the circle's $P$.*

*Proof sketch.* First, $\psi(0) = \psi(L) = 0$ satisfies the condition, so $P_D \subseteq P_\theta$, and $P_\theta \subseteq P_D^\dagger$ by Proposition 3.7. For $\varphi, \psi \in D(P_\theta)$, $\overline{\varphi(L)}\psi(L) = \overline{e^{i\theta}\varphi(0)}\,e^{i\theta}\psi(0) = \overline{\varphi(0)}\psi(0)$, so (3.2) vanishes and $P_\theta$ is symmetric. Conversely, let $\varphi \in D(P_\theta^\dagger)$. Since $P_D \subseteq P_\theta$, the adjoint $P_\theta^\dagger$ is contained in $P_D^\dagger$, so by the cited Theorem 3.8, $\varphi \in \mathcal{A}$ and $P_\theta^\dagger\varphi = -i\hbar\varphi'$. With $\psi = e_{\theta,0}$, (3.2) gives

$$0 = \langle \varphi, P_\theta\psi\rangle - \left\langle P_\theta^\dagger\varphi, \psi\right\rangle = -\frac{i\hbar}{\sqrt{L}}\left(\overline{\varphi(L)}e^{i\theta} - \overline{\varphi(0)}\right),$$

so $\varphi(L) = e^{\mathrm{i}\theta}\varphi(0)$ and $\varphi \in D(P_\theta)$. Next, $e_{\theta,k} = e^{\mathrm{i}\theta x/L} f_k$, and multiplication by $e^{\mathrm{i}\theta x/L}$, of modulus 1, is unitary, so the $e_{\theta,k}$ form a complete orthonormal basis. Each lies in $D(P_\theta)$, and $-\mathrm{i}\hbar e'_{\theta,k} = \hbar(\theta+2\pi k)e_{\theta,k}/L$. Finally, $e_{0,k} = f_k$, and $P_0$ is a self-adjoint operator with $P_0 f_k = p_k f_k$. The argument of the step "the only one" of the proof of Theorem 1.8, with $f_k$ and $p_k$, shows that $P_0 = P$. ∎

The proof is sketched because it uses the cited Theorem 3.8. There are no other self-adjoint versions.

**Proposition 3.11** (These are all · *sketched*)**.** *Every self-adjoint operator $P$ with $P_D \subseteq P$ is one of the $P_\theta$.*

*Proof sketch.* As shown before Definition 3.9, $P \subseteq P_D^\dagger$, so by the cited Theorem 3.8 $P$ is the rule $-\mathrm{i}\hbar\psi'$ on a subspace $D(P)$ of $\mathcal{A}$ that contains $D(P_D)$. Let $V \subseteq \mathbb{C}^2$ be the set of pairs $(\psi(0), \psi(L))$ with $\psi \in D(P)$, a subspace. If $\psi \in \mathcal{A}$ has its pair in $V$, some $\varphi \in D(P)$ has the same pair, and $\psi - \varphi \in D(P_D) \subseteq D(P)$. So $D(P)$ is the set of $\psi \in \mathcal{A}$ whose pair lies in $V$. Since $P$ is symmetric, every pair $(a, b)$ in $V$ has $|b| = |a|$, so $V \neq \mathbb{C}^2$. If $V = \{0\}$, then $P = P_D$, which is not self-adjoint (Proposition 3.7). So $V$ is a line, spanned by a pair $(a, b)$ with $|b| = |a| \neq 0$, that is, $b = e^{\mathrm{i}\theta}a$. Then $D(P) = D(P_\theta)$, and $P = P_\theta$. ∎

Amrein, (3.22), and Teschl, Thm 2.26 and (2.110), whose $\tilde{\theta}$ is our $\theta$, give the same family. Different $\theta$ give different eigenvalues, so the formula $-\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ does not decide the momentum. What does the choice change for a state?

**Example 3.12** (One state, two momentum distributions)**.** Take the ground state $u_1$ of the box. It lies in $D(P_D)$, so in every $D(P_\theta)$, and every $P_\theta$ acts on it by $-\mathrm{i}\hbar u'_1$.

1. *The rule.* By the observables postulate (Postulate 1.5) and Theorem 3.10, a reading of $P_\theta$ gives $\hbar(\theta + 2\pi k)/L$ with the probability $\left|\langle e_{\theta,k}, u_1\rangle\right|^2$.
2. *Read $P_0$.* Writing the sine through exponentials and integrating,

$$\langle f_k, u_1\rangle = \frac{\sqrt{2}}{L}\int_0^L \sin\frac{\pi x}{L}\, e^{-2\pi\mathrm{i}kx/L}\,\mathrm{d}x = \frac{2\sqrt{2}}{\pi(1-4k^2)}.$$

   So $P_0$ reads 0 with probability $8/\pi^2 \approx 0.811$, each of $\pm 2\pi\hbar/L$ with probability $8/9\pi^2 \approx 0.090$, each of $\pm 4\pi\hbar/L$ with probability $8/225\pi^2 \approx 0.004$, and so on.
3. *Read $P_\pi$.* Since $e_{\pi,0} = e^{\mathrm{i}\pi x/L}/\sqrt{L}$ and $e_{\pi,-1} = e^{-\mathrm{i}\pi x/L}/\sqrt{L}$,

$$u_1 = \frac{e_{\pi,0} - e_{\pi,-1}}{\mathrm{i}\sqrt{2}}.$$

   So $P_\pi$ reads $\pi\hbar/L$ or $-\pi\hbar/L$, each with probability $\frac{1}{2}$.
4. *Compare.* For $\psi \in D(P_\theta)$, symmetry gives $\langle e_{\theta,k}, P_\theta\psi\rangle = \hbar(\theta + 2\pi k)c_k/L$, with $c_k = \langle e_{\theta,k}, \psi\rangle$. By Parseval's identity, the mean of the readings is $\langle\psi, P_\theta\psi\rangle$ and their mean square is $\|P_\theta\psi\|^2$. For $u_1$ these are 0 and $\hbar^2\left\|u'_1\right\|^2 = \pi^2\hbar^2/L^2$ for every $\theta$. The two distributions have the same mean and the same mean square, and they are different.

On the ring with a flux, the flux fixes $\theta$, and the energy $P_\theta^2/2m$ depends on it: there different $\theta$ are different systems. In the box the energy $H$ is fixed (Theorem 1.8), and the $P_\theta$ are different observables of one system. Nothing in the box picks one.

A ring with a magnetic flux realises $P_\theta$. Put a flux through the ring of *Quantum Mechanics I*, §8.6: a hop to the next site gains the phase $e^{\mathrm{i}\beta}$, so $H_\beta = 2\kappa\mathbb{1} - \kappa(e^{\mathrm{i}\beta}T + e^{-\mathrm{i}\beta}T^\dagger)$. That lecture

wrote $\theta$ for $\beta$, a letter that names the boundary condition here. Change the phases of the sites: write $\psi_x = e^{\mathrm{i}\beta x}\varphi_x$ for $x = 0, \dots, N-1$. On the amplitudes $\varphi_x$, the hops inside the ring lose their phase, and the hop across from $x = N-1$ to $x = 0$ keeps the whole loop phase. In the notation of the chain, $H_\beta$ becomes the hopping with $\varphi_N = e^{-\mathrm{i}N\beta}\varphi_0$, which is the condition of $P_\theta$ for $\theta \equiv -N\beta$, modulo $2\pi$. The levels agree: with the loop phase $N\beta$ held fixed as $N \to \infty$,

$$4\kappa \sin^2 \frac{2\pi k - N\beta}{2N} \longrightarrow \frac{\hbar^2(2\pi k - N\beta)^2}{2mL^2}.$$

As $k$ runs through $\mathbb{Z}$, these limits are the numbers $(\hbar(\theta + 2\pi k)/L)^2/2m$, $k \in \mathbb{Z}$, the levels of $P_\theta^2/2m$, after the integers are relabelled. The levels alone cannot fix the sign of $\theta$. The phases of the sites do. In this sense the ring with a flux realises $P_\theta$: the change of phases is exact on the ring, and the levels converge. As for the chain of § 1.1, this is not a limit of operators.

Figure 3.1 records the ring and its boundary phase after a cut.

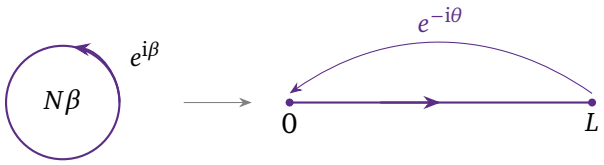



Figure 3.1: The ring's forward hops carry the phase $e^{\mathrm{i}\beta}$. After removing the phases from the interior hops, the interval has $\theta \equiv -N\beta$; the reverse boundary relation carries the phase $e^{-\mathrm{i}\theta}$.

**Self-check 3.4**

1. Find the eigenvalues of $P_\pi$. For which $\theta$ is 0 an eigenvalue of $P_\theta$?

**The physics**

A magnetic flux $\Phi$ through a ring changes the phase of a particle of charge $q$ that goes once around it, in the direction of the forward hops, by $q\Phi/\hbar$, although the field is zero on the ring itself (the Aharonov–Bohm effect). For the charge $\pm e$ of an electron or a proton this phase is $\pm 2\pi\Phi/\Phi_0$, with the flux quantum $\Phi_0 = h/e$. So the flux sets the loop phase $N\beta = q\Phi/\hbar$, and with it the boundary condition. Since the levels depend on the flux, a ring of metal about a micrometre across, cooled to a fraction of a kelvin, can carry a current in equilibrium, with no battery. Lévy, Dolan, Dunsmuir and Bouchiat detected such persistent currents in copper rings in 1990, and Bleszynski-Jayich and coworkers measured them in aluminium rings in 2009.

## 3.5 The momentum and the box's energy

For a free particle, the energy is $P^2/2m$. In the box no self-adjoint momentum squares to the energy.

**Proposition 3.13** (No momentum squares to the energy)**.** *For no $\theta$ is $P_\theta P_\theta/2m$, on the $\psi \in D(P_\theta)$ with $P_\theta\psi \in D(P_\theta)$, the box Hamiltonian $H$.*

*Proof.* The function $e_{\theta,0}$ lies in $D(P_\theta)$, and so does $P_\theta e_{\theta,0} = (\hbar\theta/L)e_{\theta,0}$. So $e_{\theta,0}$ is an eigenvector of $P_\theta P_\theta$. If $P_\theta P_\theta/2m$ were $H$, then by Theorem 1.8 $e_{\theta,0}$ would be a multiple of some

$u_n$. Both functions are continuous, and $u_n(0) = 0$, while $\left|e_{\theta,0}(0)\right| = 1/\sqrt{L}$. ∎

With the classification (Proposition 3.11), no self-adjoint version of the momentum squares to $H$. The diagonal operator $B$ with $Bu_n = (n\pi\hbar/L)u_n$, on the vectors with $\sum_n n^2 |c_n|^2 < \infty$, does satisfy $B^2/2m = H$, since $E_n = (n\pi\hbar/L)^2/2m$. But $B$ is not a version of the derivative: $-\mathrm{i}\hbar u_1'$ is a multiple of $\cos(\pi x/L)$, not of $u_1$. The box energy is built from the momentum with walls in another way: $H = P_D^\dagger P_D/2m$ (Amrein, (3.29), cited). On the twice continuously differentiable functions that vanish at the walls, this is $(-\mathrm{i}\hbar)^2\psi''/2m = -(\hbar^2/2m)\psi''$. The product $P_D^\dagger P_D$ acts on the $\psi \in D(P_D)$ with $P_D\psi \in D(P_D^\dagger) = \mathcal{A}$. This describes the domain of $H$ by derivatives.

**Theorem 3.14** (The Dirichlet domain · *cited*)**.** *The domain of the box Hamiltonian is*

$$D(H) = \left\{\psi \in \mathcal{A} :\ \psi' \in \mathcal{A},\ \psi(0) = \psi(L) = 0\right\},$$

*and* $H\psi = -(\hbar^2/2m)\,\psi''$ *for every* $\psi$ *in it.*

Teschl, (2.57), gives this domain for a self-adjoint operator that obeys the energy postulate, which is therefore $H$ by the step "the only one" of the proof of Theorem 1.8. Teta, Ex. 4.22, proves the uniqueness by another route. Both take $\hbar = 1$, $m = \frac{1}{2}$ and $L = \pi$. The Dirichlet condition $\psi(0) = \psi(L) = 0$ makes sense because a function of $\mathcal{A}$ has a value at every point. The tent $\tau$ lies in $D(P_D)$, but its derivative jumps, so $\tau' \notin \mathcal{A}$ and $\tau \notin D(H)$.

The diagonal operator $B$ above has the domain $\sum_n n^2 |c_n|^2 < \infty$. Its condition is that of a finite mean energy, $\sum_n E_n |c_n|^2 < \infty$, and the momentum with walls describes it.

**Proposition 3.15** (Finite mean energy)**.** *Let* $\psi \in L^2(0, L)$*, with* $c_n = \langle u_n, \psi\rangle$*. Then* $\sum_n E_n |c_n|^2 < \infty$ *if and only if* $\psi \in D(P_D)$*, and then*

$$\sum_{n=1}^{\infty} E_n |c_n|^2 = \frac{\|P_D\psi\|^2}{2m} = \frac{\hbar^2}{2m}\int_0^L |\psi'(x)|^2\,\mathrm{d}x.$$

*Proof.* The cosines $v_n(x) = \sqrt{2/L}\,\cos(n\pi x/L)$, $n \geq 1$, are orthonormal, by the integral of a product of two cosines, and $v_n' = -(n\pi/L)u_n$. First let $\psi \in D(P_D)$. By integration by parts (3.1), with $\psi$ vanishing at the walls,

$$\langle v_n, \psi'\rangle = -\left\langle v_n', \psi\right\rangle = \frac{n\pi}{L}\,c_n.$$

By Bessel's inequality, $\sum_n (n\pi/L)^2 |c_n|^2 \leq \|\psi'\|^2$, and $E_n = (\hbar^2/2m)(n\pi/L)^2$. So the mean energy is finite. Conversely, let $\sum_n n^2 |c_n|^2 < \infty$. Then $g = \sum_n (n\pi/L)c_n v_n$ converges, with $\|g\|^2 = \sum_n (n\pi/L)^2 |c_n|^2$. Put $\varphi(x) = \int_0^x g(t)\,\mathrm{d}t$. Then $\varphi \in \mathcal{A}$, $\varphi(0) = 0$, and $\varphi(L) = \langle 1, g\rangle = 0$, since each $v_n$ has the integral 0. So $\varphi \in D(P_D)$, and by the first part $\langle u_n, \varphi\rangle = (L/n\pi)\,\langle v_n, g\rangle = c_n$ for every $n$. By the sine basis theorem $\varphi = \psi$, so $\psi \in D(P_D)$, with $\psi' = g$ by the second fact about $\mathcal{A}$. Hence $\|\psi'\|^2 = \sum_n (n\pi/L)^2 |c_n|^2$, and multiplying by $\hbar^2/2m$ gives the formula. ∎

So the state of § 1.7, with $c_n \propto 1/n^2$, lies in $D(P_D)$ but not in $D(H)$. The state of § 2.6 that a reading of the left half leaves jumps at $L/2$, so it is not in $\mathcal{A}$, and its mean energy is infinite, as found there. The operators $B$ and $P_D$ have the same domain, and $\|B\psi\| = \|P_D\psi\|$ on it, but they are different operators.

**Self-check 3.5**

1. Does $u_1$ lie in $D(P_\theta)$ for every $\theta$? Does $P_\theta u_1$ depend on $\theta$? For which $\theta$ is $P_\theta u_1$ in $D(P_\theta)$, and what is $P_\theta P_\theta u_1$ then?

**Momentum on a circle and in a box**

*On the circle:* the momentum $P$, diagonal in the Fourier basis with the eigenvalues $2\pi\hbar k/L$, is self-adjoint, and the energy has the levels $p_k^2/2m$, the limits of the ring's levels (Proposition 3.2, § 3.1). *In the box:* the momentum with walls $P_D$ is symmetric and not self-adjoint (Propositions 3.6 and 3.7). Its self-adjoint versions are the $P_\theta$, $\psi(L) = e^{\mathrm{i}\theta}\psi(0)$, with the eigenvalues $\hbar(\theta+2\pi k)/L$, and none squares to the box's energy (Theorem 3.10 and Propositions 3.11 and 3.13). They are different observables of the box, with different readings on one state (Example 3.12). The ring with a flux realises them in the sense of § 3.4, and there $\theta$ fixes the system. *Energy:* the mean energy is finite exactly on $D(P_D)$ (Proposition 3.15).

**Summary.**

- *The momentum on a circle* is self-adjoint (Proposition 3.2), and the ring's levels tend to the circle's (§ 3.1).
- *The momentum with walls* $P_D$ is symmetric, its adjoint has every complex number as an eigenvalue, and it is not self-adjoint (Propositions 3.6 and 3.7).
- *Self-adjoint momenta in a box*: each $P_\theta$ is self-adjoint, with the eigenbasis $e_{\theta,k}$, and every self-adjoint extension of $P_D$ is one of them (Theorem 3.10 and Proposition 3.11). A flux $\beta$ per hop on the ring gives the boundary condition of $P_\theta$ with $\theta \equiv -N\beta$ (§ 3.4).
- *One state, two momentum distributions*: $u_1$ reads 0 under $P_0$ with probability $8/\pi^2$, and $\pm\pi\hbar/L$ under $P_\pi$ with probability $\frac{1}{2}$ each (Example 3.12).
- *Finite mean energy* is the domain of the momentum with walls, and the mean energy is $(\hbar^2/2m)\int|\psi'|^2$ (Proposition 3.15).
- *No momentum squares to the box's energy* (Proposition 3.13).

Lecture 4 lets the circle grow into the line, where position and momentum have no eigenvalues at all, and proves the uncertainty relation there.

**Self-check answers.**

**3.1** 1. Circle: 0 once, $4E_1$ twice, $16E_1$ twice. Box: $E_1$, $4E_1$, $9E_1$, each once.

**3.2** 1. Yes: it is continuously differentiable and vanishes at both walls. $\langle\psi, P_D\psi\rangle = -\mathrm{i}\hbar\int_0^L x(L-x)(L-2x)\,\mathrm{d}x = 0$. 2. The indicator jumps, so it is not in $\mathcal{A}$. The function $x^{3/4}$ is, with $g = \frac{3}{4}x^{-1/4}$ and $\int_0^L g^2\,\mathrm{d}x = \frac{9}{8}\sqrt{L}$; it is not in $D(P_D)$, since it is not 0 at $L$. The tent lies in $D(P_D)$.

**3.3** 1. $e^{-x/L}$, since $-\mathrm{i}\hbar(e^{-x/L})' = (\mathrm{i}\hbar/L)e^{-x/L}$. Its value at 0 is 1, so it is not in $D(P_D)$.

**3.4** 1. $(2k+1)\pi\hbar/L$ for $k \in \mathbb{Z}$, symmetric about 0. Only for $\theta = 0$.

**3.5** 1. Yes, since $u_1(0) = u_1(L) = 0$. No: every $P_\theta$ acts on it by $-\mathrm{i}\hbar u_1'$, a multiple of $\cos(\pi x/L)$, whose value at $L$ is $-1$ times its value at 0. So $P_\theta u_1 \in D(P_\theta)$ only for $\theta = \pi$, and then $P_\pi P_\pi u_1 = -\hbar^2 u_1'' = 2mE_1 u_1$.

## References

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